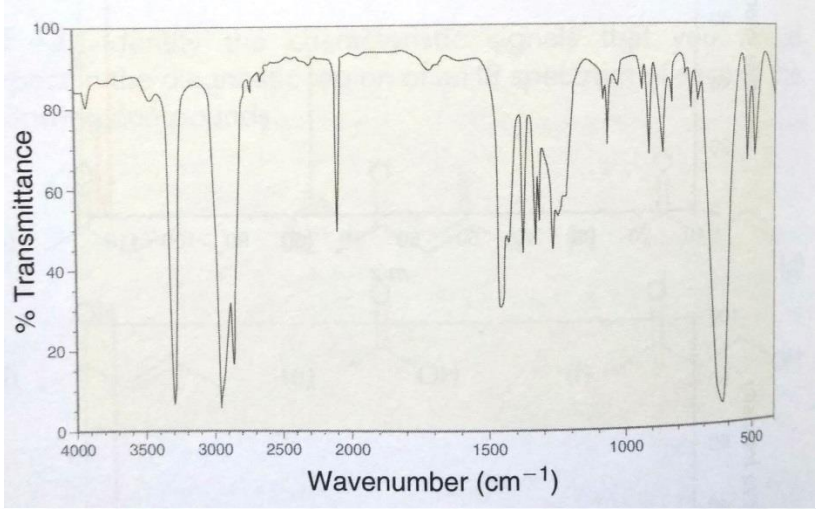
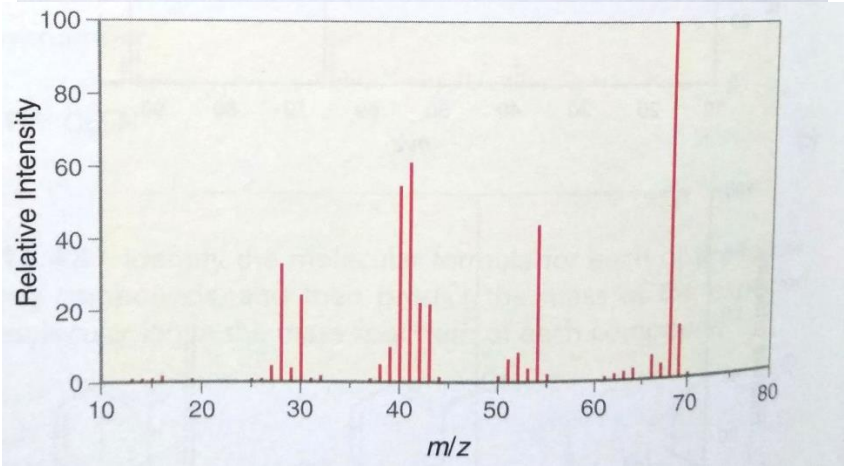
Cumulative Examination, Organic Chemistry
October 20, 2023

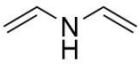
1) When treated with a strong base, 2-bromo-2,3-dimethylbutane will undergo an elimination reaction to produce two products. The choice of base (ethoxide vs. *tert.*-butoxide) will determine which of the two products predominates. Draw reaction schemes toward both products and determine how you could distinguish between the final compounds using IR spectroscopy (20 points)

2) An unknown compound gives the following IR and mass spectra. Propose at least two possible structures for the unknown compound and give a rationale for your reasoning (20 points).





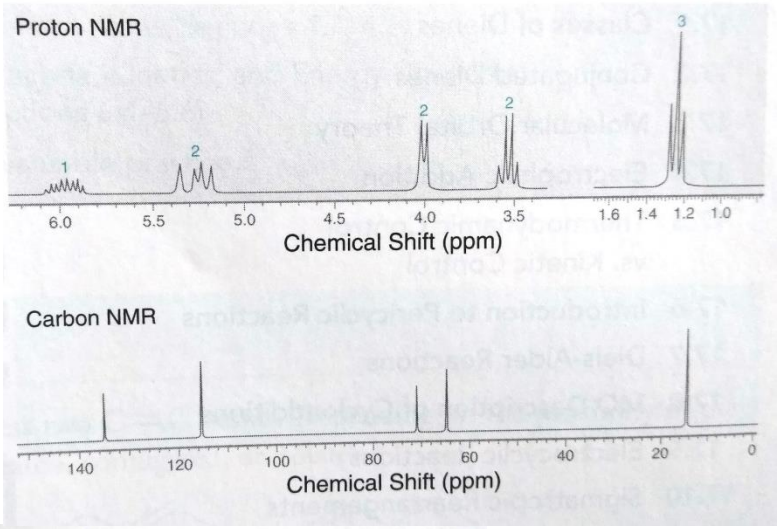
3) compare the structures of 1,4-pentadiene and divinylamine. The diene does not absorb UV light

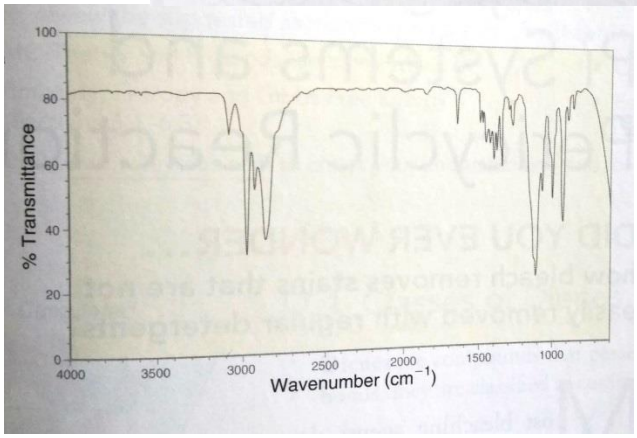


in the region between 200 and 400 nm. The amine does absorb light above 200 nm. Using this information, identify the hybridization of the N-atom in divinylamine, and explain your answer. (10 points)

4) A compound with molecular formula $C_5C_{10}O$ exhibits the following IR, ^{1}H NMR, and ^{13}C NMR spectra. The EI mass spectrum shows the data given in the table. (25 points)

m/z	Relative abund. [%]
15	23
26	20
27	61
29	92
30	20
31	45
39	47
41	**100**
45	10
57	82
58	86
86	**12**





a) Suggest a reasonable structure for the compound in question and explain your conclusion
b) Give the structure of the base peak in the mass spectrum, and a mechanism for its formation.